Eros' Metallurgy Lab (EML v4.1): Project Report

Technical project report · Version 4.1

Eros Hama Hamza · College of Engineering, Salahaddin University-Erbil · Oct 29, 2026

Abstract. Eros′ Metallurgy Lab (EML) is a browser-based workbench for materials science and mechanical design. It couples interactive specimen processing with computational thermodynamics, transformation kinetics, fracture mechanics, corrosion, manufacturing simulation and five German and European engineering codes. All 80 built-in benchmarks pass against exact solutions and measured data, and every result states its data source and model limits.

1 Overview

Eros′ Metallurgy Lab (EML) is a browser-based materials-science and mechanical-design workbench that runs entirely in one web page, with no installation. Version 4.1 combines a hands-on specimen bench, a stress analyzer, six science labs and five engineering-code calculators, and every solver is checked against textbook or measured data each time the page opens.

Table 1. EML at a glance.

Item	Count
Materials in the database (metals, alloys, polymers, ceramics, fibres, composites)	166
Automated benchmarks (all passing)	80
Science labs (microscope, thermodynamics, kinetics, fracture, corrosion, manufacturing)	6
Engineering codes implemented (EN 1993-1-1, EN 10025-2, VDI 2230, FKM, ISO 286)	5
Encyclopedia entries explaining parameters, methods and limitations	90

It is built for engineering students and young engineers who want to see why materials behave as they do, and to check a design against the methods German industry actually uses.

2 The workbench

The core of EML is a bench where you drop materials as specimens and process them the way a metallurgist would, while every property updates live.

- **Specimens:** drag any of the 166 materials onto the bench; each card shows its phase, temperature and magnetism.
- **Heat treatment:** heat, hold, quench in water, oil or air, normalise, temper; steels follow their TTT behaviour, and hardness, yield strength and toughness follow the resulting microstructure.
- **Cold work and annealing:** rolling passes raise strength by work hardening, and annealing recrystallises the grains (JMAK kinetics, Hall–Petch grain-size strengthening).
- **Alloying and composites:** drop one metal on another to make an alloy, or a fibre on a polymer to make a composite whose properties come from micromechanics.
- **Virtual testing:** tensile test, fatigue S–N curve (Basquin with surface finish and mean stress) and creep (Norton law, Larson–Miller parameter).
- **Status panels:** properties with source tags and scatter, microstructure, crystal structure, X-ray diffraction, phase diagram, TTT and Jominy curves, bonding, heating curve and real-world uses; each panel can be pinned or popped out as its own window.

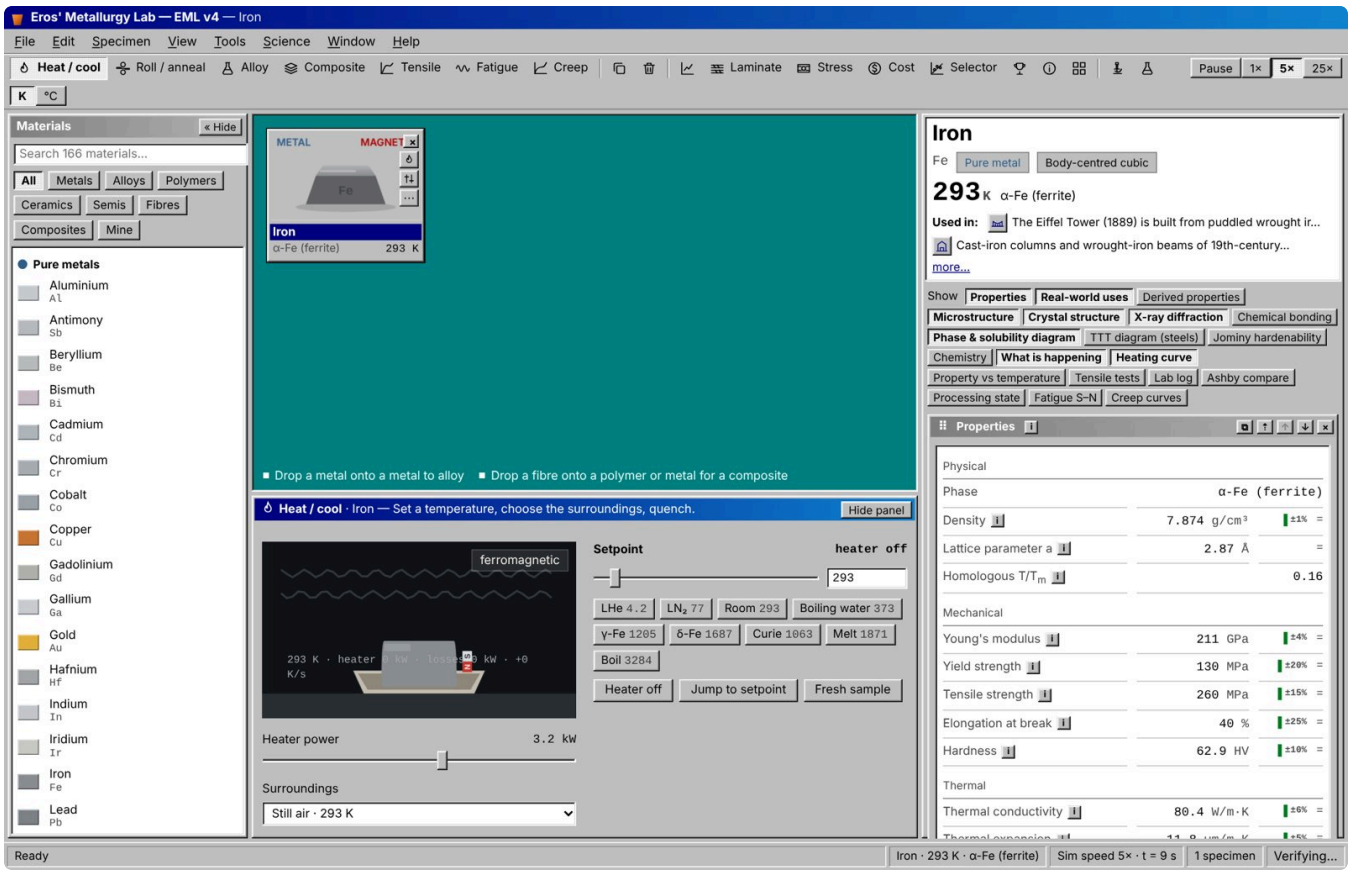


Figure 1. The workbench: materials library (left), bench and operations panel (centre), status panels (right).

3 Analysis tools

The stress analyzer starts from the shape of the part and then uses the method an engineer would use for that shape.

Table 2. Analysis tools and their methods.

Tool	What it does	Method
Beams	Rectangle, round, tube, box, I, T and C sections; four support types; point, distributed, axial and torsional loads; colour stress map, shear/moment/deflection diagrams, plastic collapse	Euler–Bernoulli beam elements (120 per beam), exact for these load cases

Tool	What it does	Method
Plates and brackets	Plate with hole, notches, stepped bar, L-bracket, lug, crack, or a shape you paint; stress concentration factor K_t	2-D plane-stress finite elements (Q4) with stress recovery and mesh convergence
Vessels and rotating parts	Thick cylinders, thin pressure vessels, rotating discs, shafts in torsion	Exact Lamé and disc solutions
Strength-of-materials toolkit	Mohr's circle with Tresca, von Mises, Rankine and Mohr–Coulomb; column buckling; thermal stress; impact loading; shaft design	Euler/Johnson, DE-Goodman
Laminate and sandwich designer	Ply stacking, ABD matrix, first-ply and progressive failure, failure envelopes, sandwich beams	Classical laminate theory with four failure criteria
Material selector	Ashby charts with performance indices and design challenges	Ashby method
Cost estimator	Material plus process cost, including composites	Process cost models

Clicking any point on a stress map sends its stresses to Mohr's circle, and every result shows a safety-factor range that reflects normal material scatter.

4 Science labs

Version 4 added six labs that compute what textbooks usually only draw, each with its method and data source stated on screen.

Table 3. Science labs added in version 4.

Lab	Highlights	Underlying science
Microscope	Etched, polarised, SEM and EBSD views at 100–1000×; pearlite colonies, martensite packets, twins, dendrites, graphite flakes; grain size (ASTM E112) and phase fractions (ASTM E562) measured on the image	Procedural metallography driven by the model state
Thermodynamics	Fe–Cr–Ni ternary sections with tie-lines and lever rule; five binary diagrams; Gibbs-energy curves; why stainless steel is stainless	CALPHAD-style regular-solution model, equilibrium by convex hull
Diffusion and kinetics	TTT and CCT diagrams for real steel compositions with hardness and Jominy curve; carburizing case depth; precipitation ageing of 6061, 7075, 2024 and 17-4 PH	Kirkaldy–Venugopalan model, Fick′s law with Tibbetts diffusivity, Shercliff–Ashby ageing
Fracture mechanics	Remaining life of a crack found in inspection, critical crack size, inspection interval	Paris law with threshold, mean-stress and K_IC effects
Corrosion	Galvanic couples with Evans diagram, pitting risk and time to perforation, stress-corrosion cracking	Mixed-potential theory, PREN, extreme-value statistics, v–K curves
Manufacturing	Casting solidification and shrinkage porosity; welding heat-affected zone and hardness traverse; metal 3-D printing residual stress and warping	Enthalpy finite differences with Niyama criterion, Rosenthal heat source, inherent-strain layer model

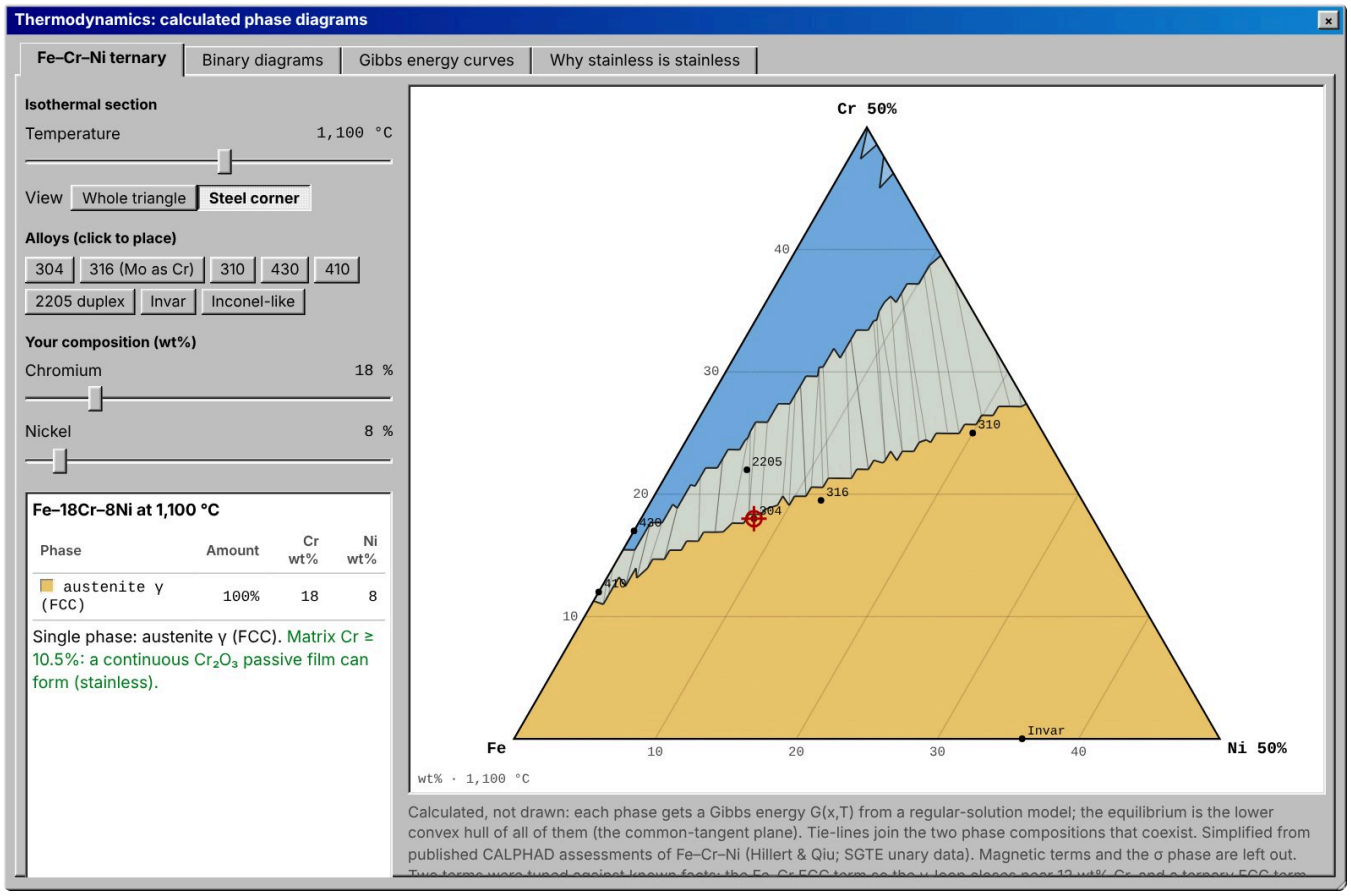


Figure 2. Calculated Fe–Cr–Ni isothermal section at 1100 °C. 304 lies in the austenite field; 2205 duplex lies in the two-phase field with tie-lines.

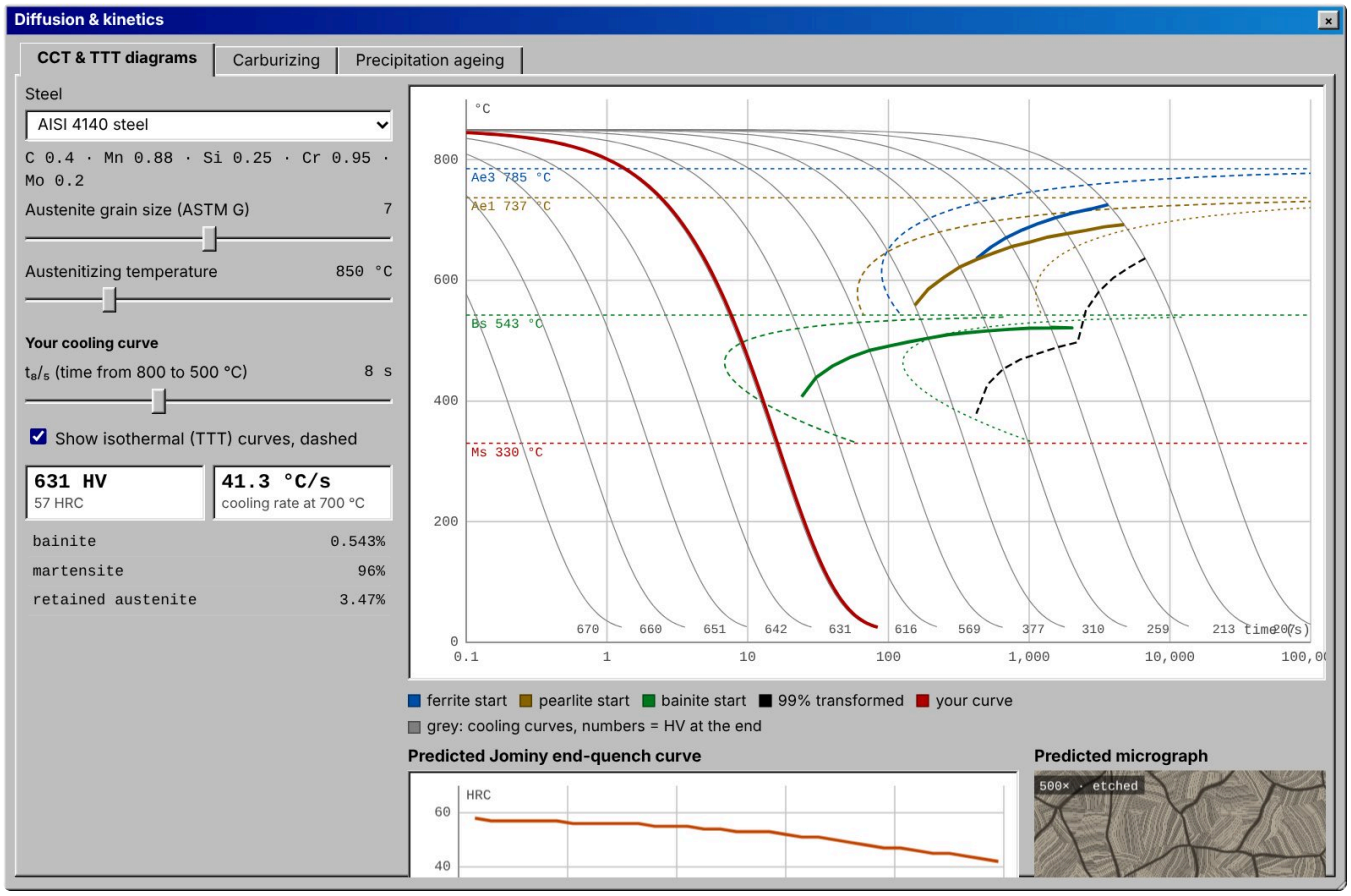


Figure 3. CCT diagram for AISI 4140 with predicted hardness, Jominy curve and micrograph.

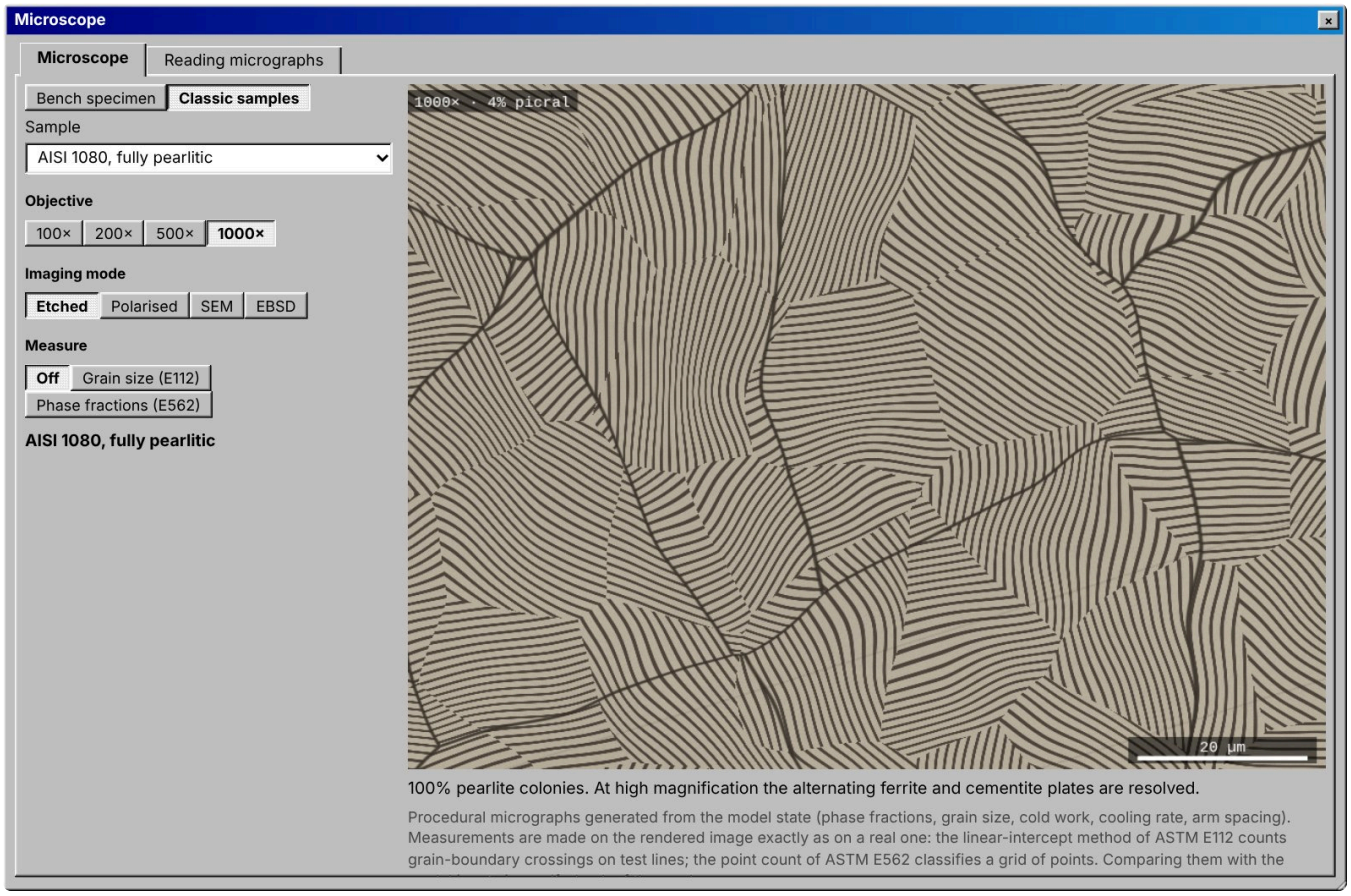


Figure 4. Microscope view of fully pearlitic AISI 1080 at 1000×, with the ferrite–cementite lamellae resolved.

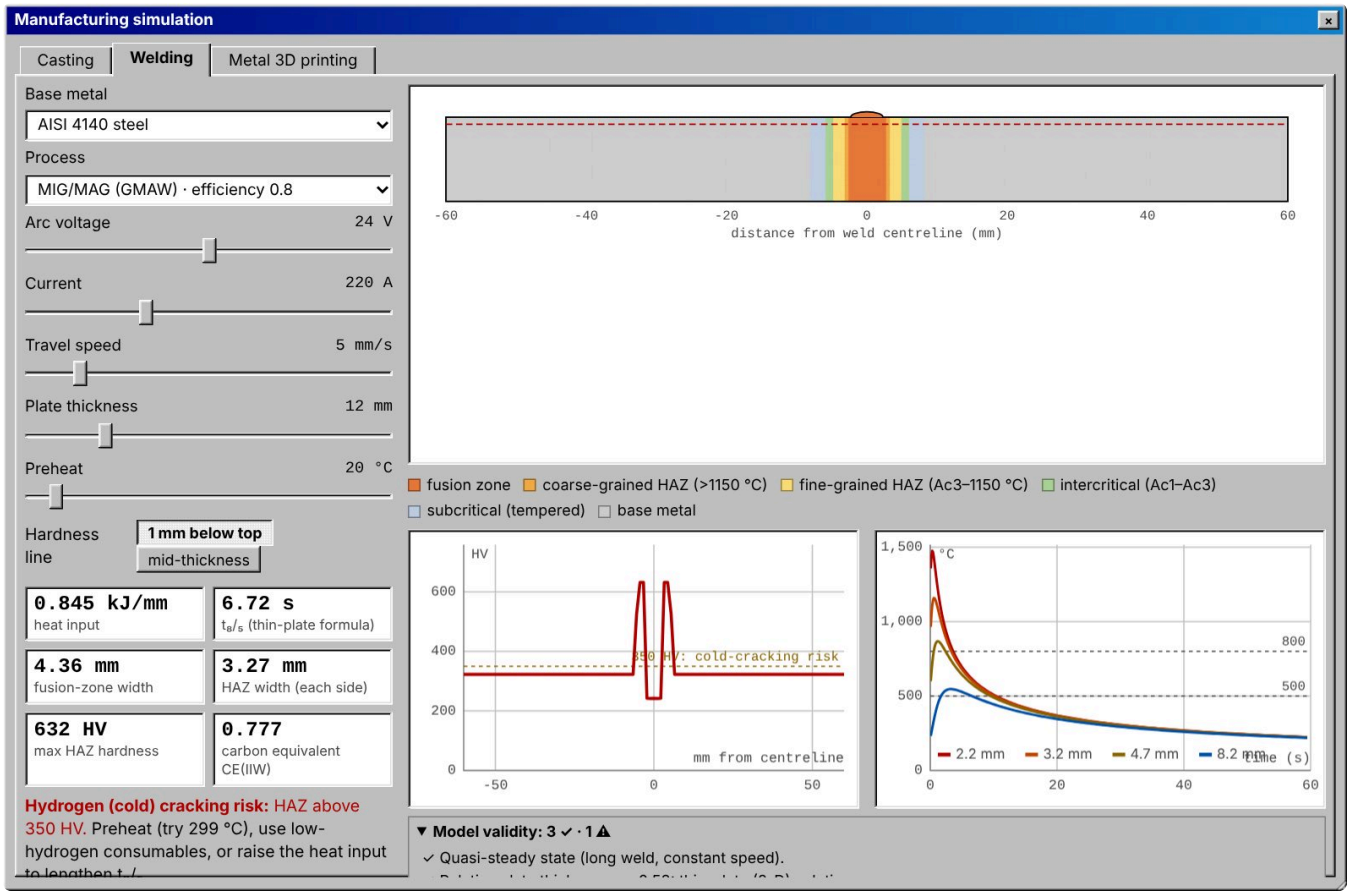


Figure 5. Welding simulation of 4140: heat-affected zones and hardness traverse.

5 Engineering codes

EML implements the core equations of five standards used daily in German and European engineering, and presents each result as a clause-referenced check table with a utilisation bar.

Table 4. Engineering codes implemented.

Standard	What EML checks
DIN EN 1993-1-1 (Eurocode 3)	IPE/HEA/HEB beams and columns: section class, bending, shear, compression, flexural and lateral-torsional buckling, Annex B interaction, deflection; German National Annex partial factors (γ_M1 = 1.1)
DIN EN 10025-2	S235–S450 yield and tensile strength by thickness, impact grades JR/J0/J2/K2, carbon equivalent

Standard	What EML checks
VDI 2230 Part 1	Bolted joints M6–M30, classes 8.8–12.9: tightening torque, load factor, preload, working stress, fatigue, surface pressure, slip (steps R4–R8)
FKM guideline	Fatigue strength of notched machine parts: support number, roughness, surface treatment, mean stress (Haigh diagram), cycles, safety factor; static check
ISO 286-1/-2	Limits and fits up to 500 mm, IT5–IT11, preferred fits H7/g6 to H7/s6, press-fit pressure, torque and shrink temperature

As an example, an unrestrained 6 m IPE 300 in S235 under 5 kN/m permanent and 8 kN/m variable load passes bending at 59% utilisation but fails lateral-torsional buckling at 112%.

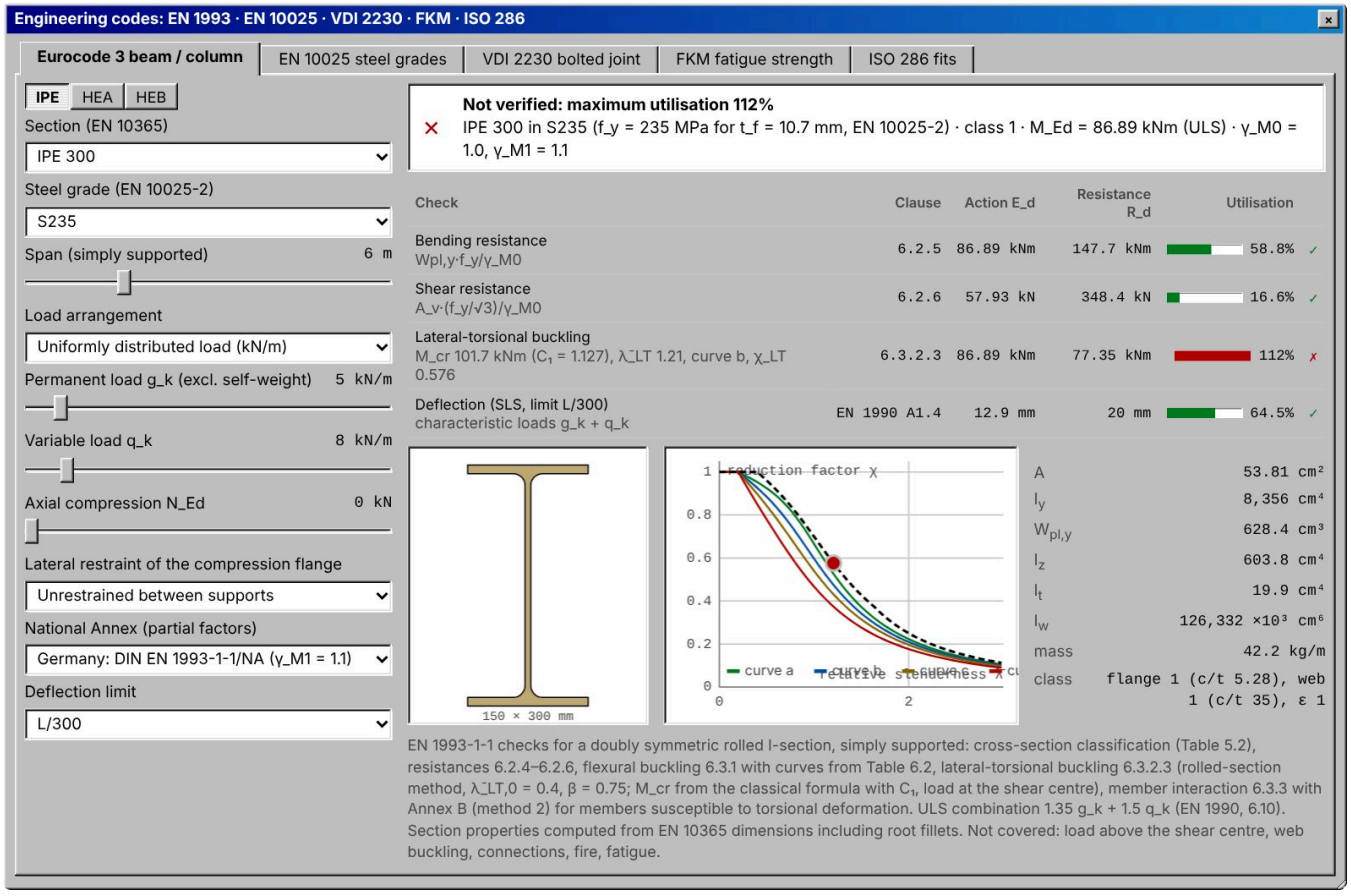


Figure 6. Eurocode 3 check of an IPE 300 beam; lateral-torsional buckling governs at 112% utilisation.

6 Verification and validation

Every result in EML can be traced and checked: all 80 benchmarks pass, and each one compares the lab's own solver with an exact textbook solution (verification) or with measured data (validation).

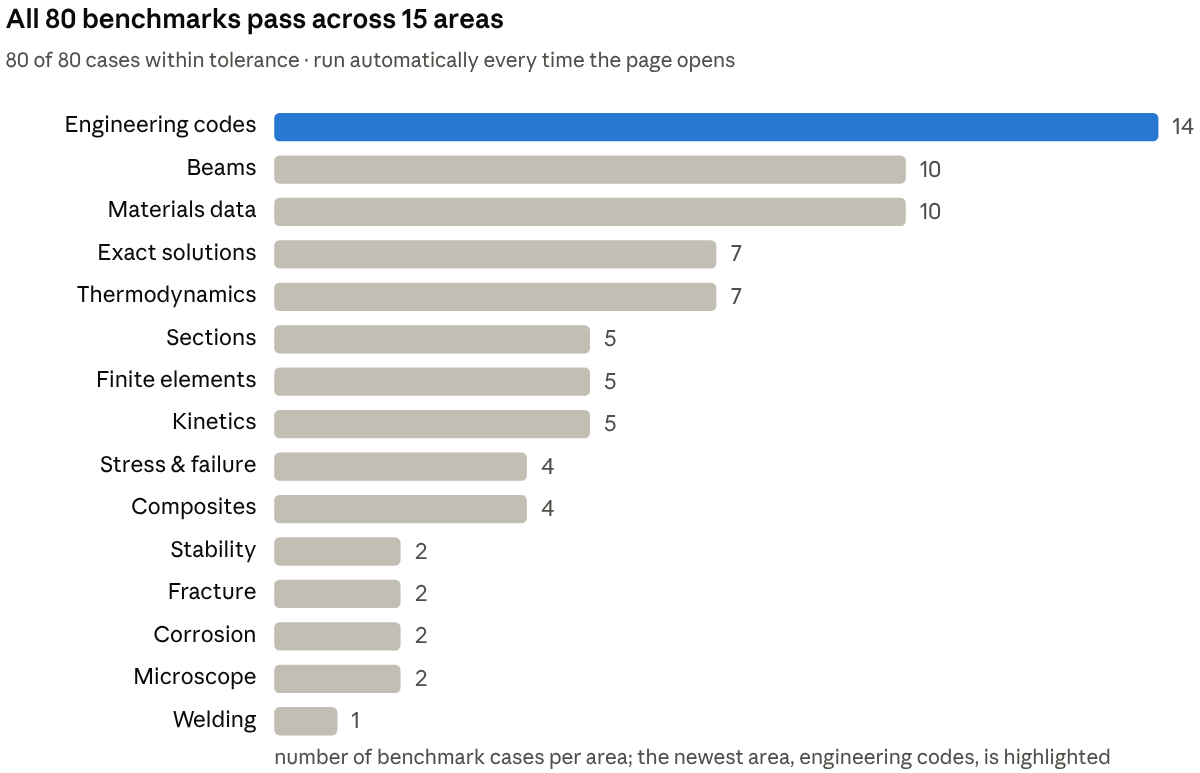


Figure 7. Benchmark cases per area, all passing.

Table 5. Selected validation cases: EML results against published reference values.

Validation case	EML result	Reference value	Source
Ag–Cu eutectic temperature	776 °C	779 °C (measured)	Massalski, Binary Alloy Phase Diagrams
Cu–Ni liquidus at 50 wt%	1306 °C	≈ 1312 °C (measured)	Massalski
2205 duplex ferrite at 1050 °C	49%	≈ 50%	Outokumpu handbook
4140 Jominy hardness at 1.5 mm	58 HRC	53–60 HRC band	ASM Handbook Vol. 4

Validation case	EML result	Reference value	Source
Cu (111) X-ray peak, Cu Kα	43.32°	43.30°	Cullity & Stock
Plate with hole K_t, fine mesh	3.57	3.59	Peterson (Heywood)
IPE 300 second moment of area	8356 cm⁴	8356 cm⁴	EN 10365 section tables
Measured grain size on rendered micrograph	G 7.68	G 7.72 (model input)	ASTM E112

Beyond the benchmarks, three features keep users honest about what the numbers mean:

- **Data provenance:** every property carries a coloured tag (literature value, design-code table, model, estimate, or user-made) and its typical batch-to-batch scatter.
- **Safety-factor ranges:** results show a 5–95% band from material scatter, not just one number.
- **Model-validity checks:** each analysis lists its assumptions and warns when they are broken, for example a coarse mesh, yielding, or a slender beam.

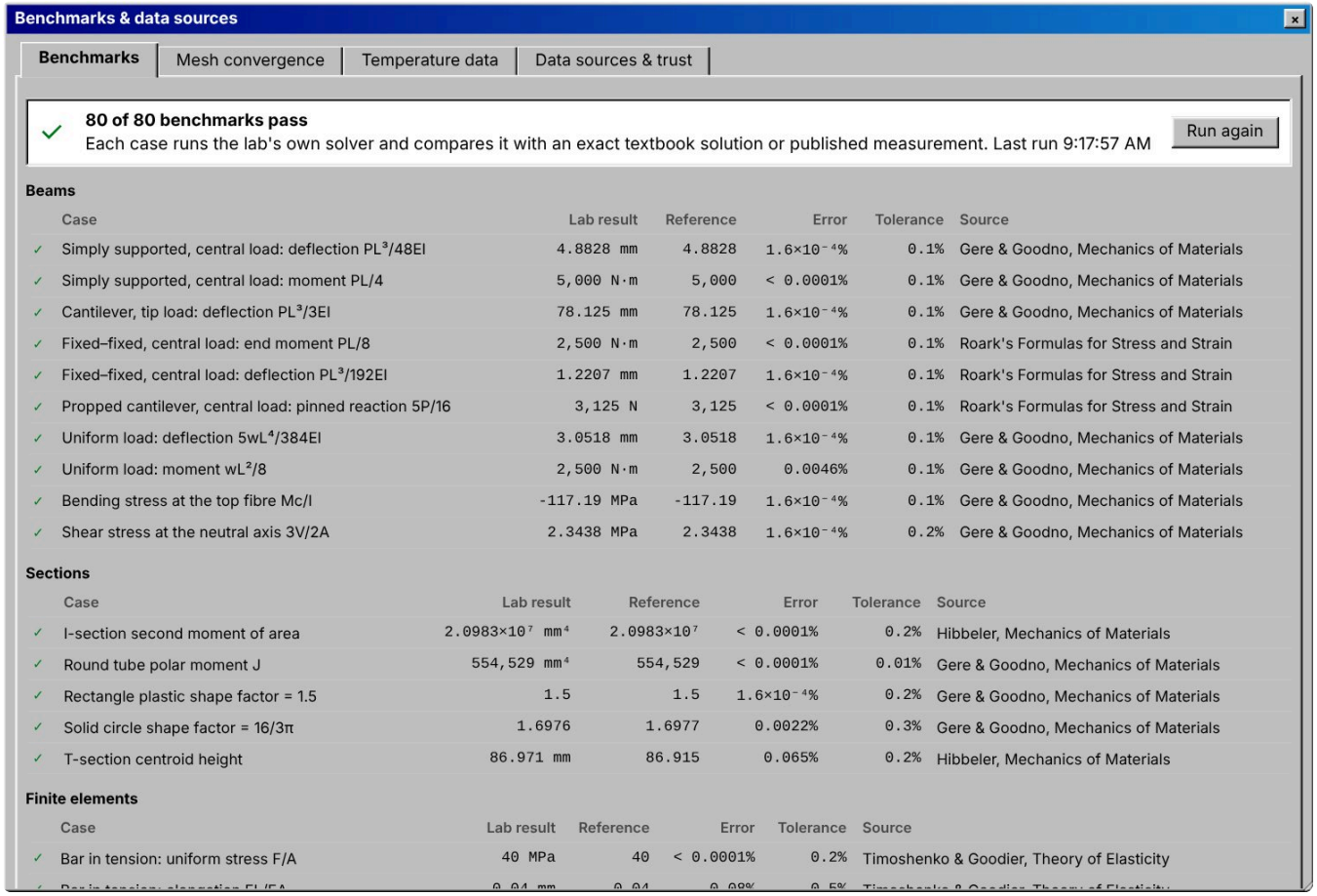


Figure 8. The benchmark page: result, reference, error, tolerance and source for every case.

7 Technology and version history

EML is a single self-contained HTML file of about 980 KB: roughly 6,900 lines of plain JavaScript in about 40 source modules, with all charts, maps, micrographs and solvers drawn on HTML canvas and no external libraries. It runs offline in any modern browser, in light and dark themes, with a classic desktop-style interface of menus, floating windows and pinnable panels.

Table 6. Version history, newest first.

Version	Added
v4.1	Engineering codes: Eurocode 3 with EN 10025, VDI 2230, FKM guideline, ISO 286; Known Limitations entries in the encyclopedia

Version	Added
v4	Microscope and new micrograph renderer; thermodynamics, kinetics, fracture, corrosion and manufacturing labs; 19 new benchmarks
v3	Benchmarks and data-sources page, data provenance tags, design-code temperature tables, safety-factor ranges, model-validity checks
v2	Shape-first stress analyzer with finite elements, strength-of-materials toolkit, laminate designer, cost estimator, real-world uses, classic desktop theme
v1	Workbench with specimens, heat treatment, alloying, composites, testing, status panels and the parameter encyclopedia

8 Limitations and future work

EML is a learning and checking tool, not certified design software, and it says so wherever it matters.

- **Property data** are typical handbook values; 27 of the 166 materials use a class estimate for fracture toughness.
- **Calibrated models:** the ageing model is fitted to one handbook peak per alloy, and two Fe–Cr–Ni parameters were tuned to known facts.
- **Indicative results:** stress-corrosion, pitting growth, 3-D printing warping and some Paris-law constants give trends and orders of magnitude only.
- **Code scope:** load above the shear centre (EC3), eccentric loading (VDI 2230), welded parts (FKM) and r/s fits above 120 mm (ISO 286) are not covered.
- **Phones** show a scaled-down desktop layout.

Planned next steps:

1. Printable calculation reports (Rechenbericht) with inputs, equations, clauses and results.
2. German-language interface with standard technical terms.

3. Evaluation of real test data: tensile tests to ISO 6892-1, hardness conversion, Charpy and S–N fits.
4. Monte Carlo reliability analysis and NAFEMS finite-element benchmarks.

References

1. EN 1993-1-1, Eurocode 3: Design of steel structures, and DIN EN 1993-1-1/NA (German National Annex).
2. EN 10025-2, Hot rolled products of structural steels.
3. VDI 2230 Part 1, Systematic calculation of highly stressed bolted joints.
4. FKM guideline, Analytical strength assessment of components.
5. ISO 286-1/-2, Geometrical product specifications: ISO code system for tolerances.
6. ASTM E112 (grain size) and ASTM E562 (point count).
7. Massalski, T. B., Binary Alloy Phase Diagrams, ASM International.
8. Kirkaldy, J. S. and Venugopalan, D. (1983); Li, M. V. et al. (1998), steel transformation kinetics.
9. Shercliff, H. R. and Ashby, M. F. (1990), process model for age hardening.
10. Gere and Goodno, Mechanics of Materials; Timoshenko and Goodier, Theory of Elasticity; Roark′s Formulas for Stress and Strain.
11. Pilkey, Peterson′s Stress Concentration Factors; Cullity and Stock, Elements of X-ray Diffraction; Callister, Materials Science and Engineering.